

Quiz 6

Imagine the chance experiment of flipping a coin three times. Let A be the event that you flip exactly two heads and let B be the event that your first flip is a head.

1. Calculate $P(B | A)$. Show your work to make your problem-solving method clear, and use English words to explain what you are doing.

I will explain how to calculate the conditional probability in several ways. You should learn all these ways to think about conditional probability.

First approach:

We use the sample space $\Omega = \{TTT, TTH, THT, THH, HTT, HTH, HHT, HHH\}$. All outcomes in this sample space are equally likely. The event A is $\{THH, HTH, HHT\}$, and the event B is $\{HTT, HTH, HHT, HHH\}$. We then have $A \cap B = \{HTH, HHT\}$.

The formula defining $P(B | A)$ is $P(B | A) = P(A \cap B) / P(A)$. Since all outcomes in Ω are equally likely, $P(A \cap B) = 2/8$ and $P(A) = 3/8$, and so $P(B | A) = 2/3$.

Second approach:

To calculate $P(B | A)$, we reduce the sample space from Ω to A. All outcomes are still equally likely. Two of the three outcomes in the reduced sample space A are in B, and so $P(B | A) = 2/3$.

Third approach:

We use Chapter 2 skills to calculate the probabilities without listing equally likely outcomes.

$P(A) = C_{3,2} (1/2)^2 (1/2)^1 = 3/8$ (binomial probability to count “successes”: H flips)

$P(A \cap B)$: This event is “first flip is H” and “exactly one H in the other two flips.” Those events are independent, and so $P(A \cap B) = (\text{probability first flip is H}) (\text{probability of exactly one H in other two flips}) = (1/2)(2/4) = 2/8$.

Therefore, $P(B | A) = P(A \cap B) / P(A) = 2/3$.

2. Calculate $P(A | B)$. As above, show your work, and explain what you are doing using English words.

I again offer several solution methods.

First approach:

As in the first approach above, we use the sample space $\Omega = \{TTT, TTH, THT, THH, HTT, HTH, HHT, HHH\}$. All outcomes in this sample space are equally likely. The event A is $\{THH, HTH, HHT\}$, and the event B is $\{HTT, HTH, HHT, HHH\}$. We then have $A \cap B = \{HTH, HHT\}$.

The formula defining $P(A|B)$ is $P(A|B) = P(A \cap B)/P(B)$. Since all outcomes in Ω are equally likely, $P(A \cap B) = 2/8$ and $P(B) = 4/8$, and so $P(B|A) = 2/4 = 1/2$.

Second approach:

To calculate $P(A|B)$, we reduce the sample space from Ω to B . All outcomes are still equally likely. Two of the four outcomes in the reduced sample space B are in A , and so $P(B|A) = 2/4$.

Third approach:

We use Chapter 2 skills to calculate the probabilities without listing equally likely outcomes.

$P(B) = 1/2$.

$P(A \cap B)$: This event is “first flip is H” and “exactly one H in the other two flips.” Those events are independent, and so $P(A \cap B) = (\text{probability first flip is H}) (\text{probability of exactly one H in other two flips}) = (1/2)(2/4) = 2/8$.

Therefore, $P(A|B) = P(A \cap B)/P(B) = 4/8$.

Fourth approach:

$P(A|B)$ is the probability of exactly two heads given that the first flip is heads. This conditional probability is the probability of flipping exactly one head in the last two flips. That probability is the binomial probability $C_{2,1}(1/2)^1(1/2)^1 = (2)(1/2)^2 = 2/4$.